

Assignment 3

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1 Introduction

The objective of this assignment is to simulate the diffusion of information through a social network where different types of actors coexist. The network represents interactions between anime entries extracted from the MyAnimeList dataset. Each node represents an anime, while edges represent similarity relationships based on shared genres.

The simulation models the spread of different types of information: a reliable message propagated by verifiers, a mutated message generated by unintentional spreaders, and false information introduced by social bots.

The main goal is to study how heterogeneous behaviors influence information diffusion and how misinformation can emerge from network dynamics.

2 Network Construction

The network was generated from the filtered anime dataset used in the previous assignments.

Each anime is represented by its identifier (`mal_id`) and title. An undirected edge is created between two nodes when they share at least two genres.

The graph construction rule is:

- Nodes: anime entries.
- Edges: two anime share at least two genres.
- Edge type: undirected.

The final network contains:

Metric	Value
Nodes	3579
Edges	780393
Average degree	436.10
Density	0.1219

Table 1: Main characteristics of the anime similarity network.

A fixed random seed (42) was used to guarantee reproducibility.

3 Actor Roles

The simulation considers four different types of nodes.

3.1 Verifiers

Verifiers represent reliable sources of information. They always start with the correct message:

$$M_{true} = Low$$

They act as trusted nodes and never change their state during the simulation.

3.2 Unintentional Spreaders

Unintentional spreaders represent users that accidentally propagate modified information. If they observe enough neighbors supporting the true message, they transform it into a mutated version:

$$M_{mutated} = Moderate$$

This models the degradation of information caused by imperfect transmission.

3.3 Normal Nodes

Normal nodes adopt the message supported by the majority of their neighbors. The adoption rule is based on a threshold:

$$\frac{neighborssupportingmessage}{totalneighbors} \geq threshold$$

If no message reaches the threshold, the node remains inactive.

3.4 Social Bots

In the second part of the experiment, malicious social bots are introduced. Bots are selected among the nodes with the highest betweenness centrality, because highly central nodes can influence a larger portion of the network. Bots always spread:

$$M_{fake} = Fake$$

and can override normal diffusion processes.

4 Diffusion Model

The simulation evolves in discrete steps.

At every iteration:

1. Verifiers maintain the true message.
2. Unintentional nodes can mutate reliable information.
3. Normal nodes update their state according to neighbor influence.
4. Bots spread fake information when activated.

The process stops when the network reaches a stable configuration or after 20 iterations.

5 Experimental Setup

The first experiment was performed with:

- 200 verifier nodes;
- 10 unintentional spreaders;
- threshold = 0.01.

The second experiment introduced:

- 3 social bots;
- bot selection based on betweenness centrality;
- threshold = 0.3.

The different thresholds were chosen because bots represent a stronger disruptive mechanism and require a stricter adoption condition.

6 Results

6.1 Diffusion Without Bots

The final distribution of messages without social bots was:

Message	Percentage
True	80.53%
Mutated	0.22%
None	19.25%

Table 2: Final message distribution without social bots.

The results show that the presence of 200 verifier nodes allows the correct message to reach most of the network. Only a small fraction of nodes adopts the mutated message generated by unintentional spreaders.

A significant percentage of nodes remains inactive because the adoption threshold is not reached by their neighbors.

6.2 Diffusion With Bots

After introducing three social bots selected using betweenness centrality, the final state becomes:

Message	Percentage
True	5.62%
Mutated	0.00%
Fake	0.00%
None	94.38%

Table 3: Final message distribution with social bots.

The introduction of bots produces a strong reduction of the propagation of the reliable message. However, in this specific simulation configuration the fake message does not spread widely because the adoption mechanism requires additional neighbor support and the simulation terminates when the system reaches a stable state.

This result shows that highly central malicious nodes can disrupt information diffusion even when they do not completely dominate the network.

7 Phase Transition Analysis

To analyze the effect of accidental misinformation propagation, the number of unintentional nodes was progressively increased from 0 to 40.

For each configuration, 10 simulations were performed and the average fraction of mutated messages was measured.

The experiment studies whether the network presents a transition point where small increases in unreliable actors produce a large increase in misinformation.

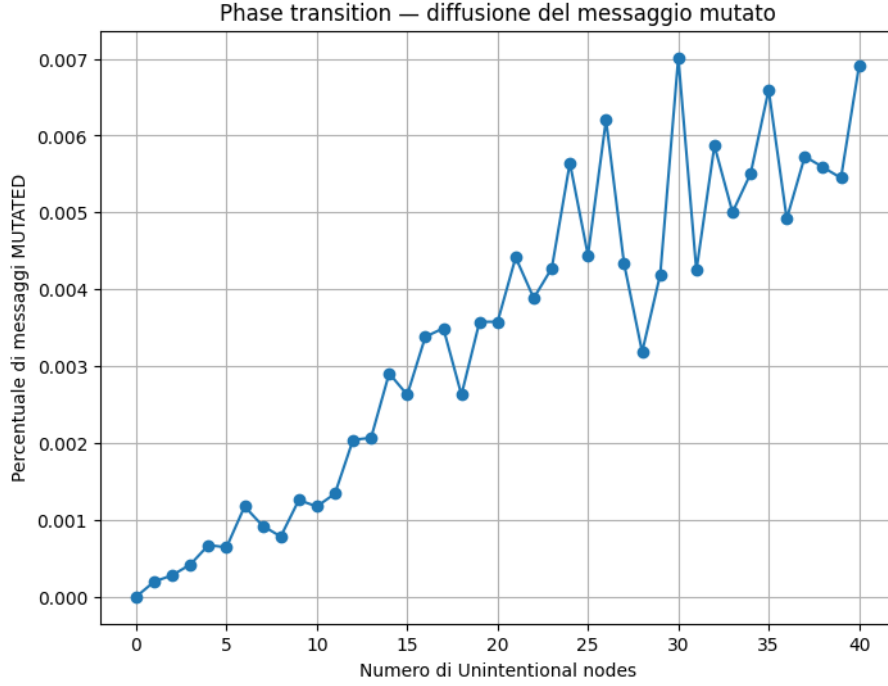


Figure 1: Percentage of mutated messages as a function of unintentional nodes.

The results show that increasing the number of unintentional spreaders increases the probability of information degradation.

8 Network Visualization

The final network states were visualized using a spring layout.

Node colors represent:

- Green: verifier nodes.
- Yellow: unintentional spreaders.
- Red: social bots.
- Blue: true information.
- Orange: mutated information.
- Purple: fake information.
- Gray: inactive nodes.

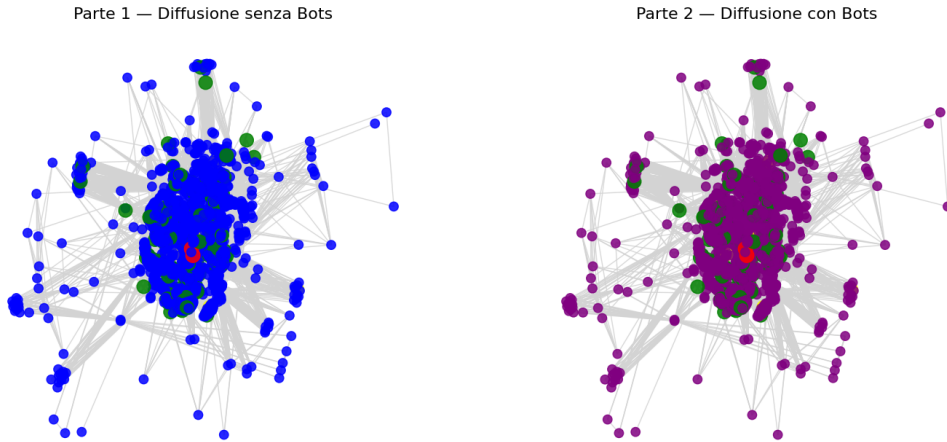


Figure 2: Comparison between diffusion without bots and with bots.

9 Technical Decisions

Several design choices were made:

- A fixed random seed was used to ensure reproducibility.
- Betweenness centrality was selected for bot placement because it identifies nodes acting as bridges in information flows.
- A threshold-based adoption model was used to simulate social influence.
- Multiple message states were introduced to model gradual information degradation instead of a simple true/false scenario.

10 Conclusion

The simulation demonstrates that network structure and actor heterogeneity strongly influence information diffusion.

Reliable sources can propagate correct information, but unintentional spreaders introduce degradation effects. Furthermore, a small number of strategically positioned bots can significantly alter the final state of the network.

The experiment highlights the importance of considering both topology and node behavior when analyzing information propagation in social networks.